THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 10: **Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Rays parallel to the principal axis, after reflection from a concave mirror, pass through the: 1

(a) centre of the object **(b)** pole
(c) principal focus **(d)** none

2. Which lens can be used as a magnifying glass? 1

(a) concave lens **(b)** thick prism
(c) convex lens of small focal length **(d)** plane glass

3. The minimum height of a plane mirror to see one's full image is: 1

(a) equal to the height **(b)** double
(c) half the height of the person **(d)** one-third

4. A ray passing through the focus of a concave mirror after reflection travels: 1

(a) parallel to the principal axis **(b)** back along the same path
(c) perpendicular to the axis **(d)** through the pole

5. A convex lens held close to a printed page shows letters: 1

(a) inverted and smaller **(b)** erect and enlarged
(c) no change **(d)** invisible

6. The number of images formed by two plane mirrors inclined at 45 degrees is: 1

(a) 8 **(b)** 6 **(c)** 7 **(d)** 3

7. A ray passing through the centre of curvature after reflection: 1

(a) goes to the pole **(b)** goes parallel to the axis
(c) retraces its path **(d)** passes through the focus

8. A convex lens held far from the page gives an image that is: 1

(a) no image **(b)** same **(c)** inverted **(d)** erect

9. The number of images formed by two parallel plane mirrors is: 1

(a) infinite **(b)** 2 **(c)** zero **(d)** 1

10. The image formed by a convex mirror is always: 1

(a) virtual and enlarged **(b)** real and enlarged
(c) real and inverted **(d)** virtual, erect and diminished

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Calculate the number of images formed if mirrors are inclined at 72 degrees. 2

12. Differentiate between a real and a virtual focus. 2

13. Why do we use mirrors with a concave reflecting surface in solar cookers? 2

14. Why does the writing on an ambulance appear inverted on the front? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why the image of an object kept very close to a concave mirror is magnified and erect, with a ray diagram description. 3

16. Why do streetlights and headlights use reflectors? 3

17. Describe an experiment to show that the angle of incidence equals the angle of reflection. 3

OR

State and explain the rules for the ray diagrams for a convex lens. 3

18. How can you get a real image of a candle flame using a convex lens? State the arrangement and the nature of the image. 3

OR

Why do the images of the Sun and the Moon formed by a convex lens appear as a small spot? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the image formation by a concave mirror for six positions of the object. Tabulate position, nature and size. 5

OR

Describe how a pinhole camera forms an image, and compare it with a camera that uses a lens. 5

20. Explain how rays are traced through convex and concave lenses and describe the nature of the image when an object is placed beyond 2F, between F and 2F and within F for a convex lens. 5

OR

Explain what is meant by magnification with the example of a magnifying glass, and describe how its use can be explored with a fair test. 5

END OF QUESTION PAPER — Total: 40 Marks